

Banking Account Management System (BAMS)

Comprehensive Project Report

Version 1.0 | June 2026

Submitted by: Koushik & Team

Table of Contents

1. Abstract	3
2. Project Objectives	4
3. System Architecture	5
4. Database Schema	6
5. Module Description	7
6. Technology Stack	12
7. Testing & Quality Assurance	13
8. Deployment & Hosting	14
9. Conclusion	15

1. Abstract

The Banking Account Management System (BAMS) is a state-of-the-art web application designed to digitize traditional banking operations. The system provides a seamless experience for customers to manage their finances, including transfers, loans, and investments, while offering a powerful administrative interface for bank officials to monitor transactions and manage accounts. Built with modern web technologies, BAMS prioritizes security, responsiveness, and user engagement.

2. Project Objectives

The primary objectives of the BAMS project are:

- To provide a secure and user-friendly digital banking platform.
- To implement modular banking operations like Fund Transfer, Loan Application, and Card Management.
- To build a robust Admin Panel for centralized monitoring and customer support.
- To ensure data integrity through a strictly defined SQL database schema.
- To deliver high-performance visual analytics for financial tracking.

3. System Architecture

BAMS follows a standard 3-tier architecture: 1. Presentation Layer: Built using HTML5, Vanilla CSS3, and JavaScript (ES6). 2. Business Logic Layer: Implemented with Spring Boot 3 framework for RESTful APIs. 3. Data Layer: Utilizes JPA/Hibernate for object-relational mapping with a relational database.

4. Database Schema

The core database entities include:

Table Name	Description
Users	Stores authentication and personal profile data
Accounts	Manages balances and account status
Transactions	Logs all financial movements (Audit Trail)
Loans	Tracks application, tenure, and repayment status
Cards	Virtual card settings, limits, and blocking status
Deposits	Investment details for Fixed Deposits

5. Module Description

5.1 Customer Modules

- Dashboard: Real-time asset overview and spending analytics.
- Transfer: NEFT, RTGS, and IMPS support with OTP security.
- Cards: 3D flip animation for virtual cards with limit management.
- Loans: Application hub with integrated EMI calculator.

5.2 Admin Modules

- Admin Dashboard: System-wide KPI tracking (Users, Revenue).
- Account Management: Activate, deactivate, and search customers.
- Global Audit: Real-time monitoring of all financial activity.

9. Conclusion

BAMS successfully fulfills the requirement for a modern digital banking solution. By integrating advanced UI features with a secure backend architecture, the project provides a comprehensive template for educational and commercial banking systems.